A THIRD-ORDER ALGEBRAIC DIFFERENTIAL EQUATION AND ALL-ORDER $\mathbb{C}(q)$-LINEAR OBSTRUCTION FOR $\eta(\tau)^{24}$

ARGUS OPEN RESEARCH PROJECT

Abstract. In connection with AIM Problem 1.2 on differential equations for η-products, we record an explicit third-order nonlinear polynomial differential equation with rational coefficients satisfied by the discriminant modular form $y(q) = \Delta(\tau) = \eta(\tau)^{24}$ under $D = q\frac{d}{dq}$, and prove that $y(q)$ satisfies no nonzero finite-order homogeneous linear differential equation over $\mathbb{C}(q)$.

1. Main Theorem

Let $q = e^{2\pi i\tau}$ ($\operatorname{Im}\tau > 0$), $D = q\frac{d}{dq} = \frac{1}{2\pi i}\frac{d}{d\tau}$, and $y(q) = \eta(\tau)^{24} = q\prod_{n=1}^{\infty}(1-q^n)^{24}$.

Theorem 1.1. *The function $y(q) = \eta(\tau)^{24}$ satisfies:*

(1) Third-order nonlinear algebraic ODE: *eliminating the Eisenstein series E_2, E_4, E_6 from Ramanujan's differential system $DE_2 = (E_2^2 - E_4)/12$, $DE_4 = (E_2E_4 - E_6)/3$, $DE_6 = (E_2E_6 - E_4^2)/2$, together with $Dy = E_2y$ and $1728y = E_4^3 - E_6^2$, yields the exact homogeneous degree-12 polynomial identity in (y, Dy, D^2y, D^3y):*

$$\left(24\,y^2D^3y - 90\,y\,Dy\,D^2y + 65\,(Dy)^3\right)^2 - \left(12\,y\,D^2y - 13\,(Dy)^2\right)^3 - 1728\,y^6 \;=\; 0.$$

(2) All-order homogeneous $\mathbb{C}(q)$-linear obstruction: *for every integer $m \geq 0$ and every tuple of rational functions $(a_0(q), \ldots, a_m(q)) \in \mathbb{C}(q)^{m+1} \setminus \{0\}$, one has $\sum_{j=0}^{m} a_j(q)D^jy(q) \not\equiv 0$.*

Proof of (2). By clearing denominators, assume $a_j(q) \in \mathbb{C}[q]$ are not all zero. Select a root of unity $\zeta = e^{2\pi ic/d}$ at which not all $a_j(\zeta)$ vanish (or after dividing out $(q-\zeta)^r$). Under the modular transformation $\tau = c/d + it$ as $t \to 0^+$, $\eta(\tau)^{24} = (dt)^{-12}e^{-2\pi/(d^2t)}(1 + O(e^{-2\pi/(d^2t)}))$ vanishes to infinite order along the radial ray $q = e^{-2\pi t}\zeta \to \zeta$, so $\lim_{t\to 0^+}(q-\zeta)^{-k}D^jy(q) = 0$ for all $j, k \geq 0$, contradicting the leading nonvanishing Taylor term of $\sum_{j=0}^{m} a_j(q)D^jy(q)$ at $q = \zeta$. □

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